

MOLOKAI, HI, USA

WMO: 911860

Lat: 21.155N

Lon: 157.096W

Elev: 135

StdP: 99.71

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22534

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.1	16.4	9.7	7.6	21.3	11.5	8.6	20.1	11.5	23.2	10.8	23.3	0.9	50	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.5	30.9	23.6	30.1	23.3	29.4	22.9	24.8	28.4	24.3	28.0	24.0	27.6	5.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.9	26.3	23.1	18.2	26.1	22.7	17.7	25.9	75.7	28.6	73.8	27.9	72.2	27.7	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.7	9.0	DB	13.0	32.2	1.0	1.6	12.2	33.4	11.6	34.4	11.1	35.3	10.3	36.5	(4)
(5)			WB	11.5	25.4	1.3	1.5	10.6	26.5	9.8	27.4	9.1	28.2	8.2	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	21.7	21.7	22.0	22.8	23.8	25.0	25.8	26.1	25.9	25.3	24.0	22.7	(6)
(7)		DBStd	2.06	1.56	1.42	1.49	1.27	1.18	0.89	1.09	0.96	1.11	1.18	1.26	1.47	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	1	0	0	1	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5079	363	327	372	385	427	451	488	500	478	474	421	393	(10)
(11)		CDD18.3	2038	104	94	114	135	169	201	230	242	228	216	171	134	(11)
(12)		CDH23.3	11527	329	272	369	504	896	1248	1617	1823	1729	1441	821	479	(12)
(13)		CDH26.7	2083	17	10	22	29	106	229	369	453	427	293	97	32	(13)
(14)	Wind	WSAvg	4.9	4.2	4.4	4.6	5.2	4.7	5.5	5.8	5.6	4.8	4.6	4.4	(14)	
(15)	Precipitation	PrecAvg	740	124	98	76	60	42	16	17	14	18	56	83	119	(15)
(16)		PrecMax	1117	304	363	366	183	181	48	39	44	55	188	303	300	(16)
(17)		PrecMin	328	16	7	4	10	0	0	0	1	2	1	5	0	(17)
(18)		PrecStd	254	93	78	75	52	45	14	10	13	12	55	87	80	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.5	27.8	28.8	28.5	29.6	30.5	31.4	31.5	31.7	31.2	29.7	29.0	(19)
(20)			MCWB	19.9	20.8	20.9	20.9	22.6	23.2	23.9	24.2	24.1	23.3	22.4	21.6	(20)
(21)		2%	DB	27.1	26.9	27.3	27.4	28.4	29.2	30.3	30.5	30.4	30.0	28.6	27.4	(21)
(22)			MCWB	20.3	20.7	20.9	21.1	22.0	22.6	23.6	23.7	23.6	23.2	22.2	21.0	(22)
(23)		5%	DB	26.1	26.1	26.3	26.9	27.7	28.7	29.3	29.6	29.6	29.0	27.8	26.6	(23)
(24)			MCWB	20.3	20.4	20.6	20.9	21.7	22.3	23.1	23.3	23.2	22.9	21.9	20.8	(24)
(25)		10%	DB	25.2	25.1	25.6	26.1	27.2	27.9	28.7	28.9	28.9	28.2	27.2	26.0	(25)
(26)			MCWB	20.0	20.1	20.3	20.6	21.5	21.9	22.7	22.9	22.9	22.6	21.7	20.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	23.2	23.0	23.5	23.9	24.1	25.1	25.3	25.5	25.1	24.2	23.2	(27)
(28)			MADB	24.8	25.1	26.0	25.8	27.1	28.3	28.8	29.3	29.2	28.0	27.1	26.2	(28)
(29)		2%	WB	22.1	22.5	22.2	22.3	23.2	23.5	24.3	24.6	24.7	24.3	23.5	22.4	(29)
(30)			MADB	24.4	24.6	25.3	25.7	26.7	27.7	28.3	28.5	28.3	27.6	26.5	25.4	(30)
(31)		5%	WB	21.6	21.8	21.6	21.8	22.7	22.9	23.8	24.1	24.1	23.9	22.9	21.9	(31)
(32)			MADB	24.2	24.2	24.7	25.3	26.3	27.2	27.9	27.9	27.8	27.3	26.1	24.9	(32)
(33)		10%	WB	21.0	21.0	21.0	21.3	22.1	22.5	23.3	23.7	23.6	23.4	22.4	21.4	(33)
(34)			MADB	23.9	23.9	24.2	24.8	25.9	26.8	27.3	27.5	27.4	26.9	25.7	24.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	7.1	6.9	6.6	7.1	6.7	6.4	6.5	6.9	6.8	6.4	6.8	(35)
(36)			MCDBR	8.5	8.3	8.2	7.4	7.9	7.4	7.0	7.1	7.5	7.6	7.3	7.8	(36)
(37)			MCWBR	4.4	4.4	4.2	3.7	3.8	3.4	3.2	3.1	3.4	3.5	3.6	4.1	(37)
(38)		5% WB	MADB	7.4	7.1	7.2	6.9	7.3	6.9	6.5	6.7	7.1	7.2	6.7	6.7	(38)
(39)	MCWBR		4.4	4.1	4.1	3.7	3.8	3.4	3.3	3.2	3.6	3.6	3.7	3.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.347	0.353	0.361	0.380	0.375	0.362	0.361	0.360	0.352	0.360	0.359	0.344	(40)	
(41)		taud	2.625	2.597	2.584	2.508	2.552	2.595	2.609	2.613	2.641	2.626	2.626	2.651	(41)	
(42)		Ebn at Noon	931	943	944	923	917	921	923	930	939	924	911	922	(42)	
(43)		Edh at Noon	88	95	100	110	104	99	97	97	93	91	87	83	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.45	5.01	5.70	6.46	6.92	7.20	7.08	6.83	6.40	5.39	4.54	4.10	(44)	
(45)		RadStd	0.19	0.33	0.56	0.38	0.37	0.20	0.16	0.27	0.19	0.15	0.11	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+91	(47)	

Nomenclature: See separate page